1 When a light plastic ball is placed in a vertical column of moving air, the ball becomes stationary at a height *h*, as shown in Fig. 1.1.

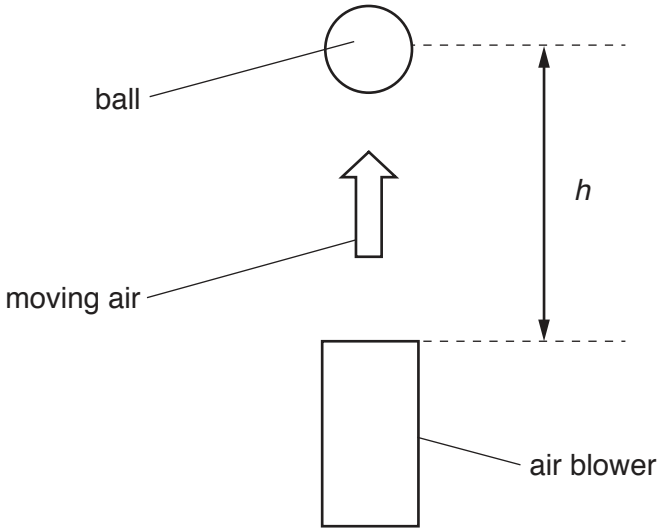


Fig. 1.1

A student is using an air blower to create the vertical column of moving air. The student connects the motor of the air blower to a d.c. power supply.

It is suggested that the relationship between the radius *r* of the ball and *h* is

$$\frac{4\pi r^3 gh}{3} = PK$$

where *g* is the acceleration of free fall, *P* is the power of the motor and *K* is a constant.

Design a laboratory experiment to test the relationship between *r* and *h*.
Explain how your results could be used to determine a value for *K*.

You should draw a diagram, on page 3, showing the arrangement of your equipment. In your account you should pay particular attention to:

- the procedure to be followed
- the measurements to be taken
- the control of variables
- the analysis of the data
- any safety precautions to be taken.

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